Rate My Professors Reviews

Idea: Do students rate professors highly because they teach well, or because they're more lenient? Study RMP reviews from the University of Rochester (and possibly nearby schools) and separate the effect of leniency from the effect of teaching quality.

Data: Collecting U of R reviews including variables like department, overall quality rating, difficulty variable, review text, etc.

- If scraping goes against RMP's terms of use, could use Mendeley's RateMyProfessors dataset with millions of reviews

Features (feature engineering):

- *From text*: A leniency score from a dictionary (easy, curve, lenient, extra credit), a teaching quality score (clear, explains, organized, helpful, engaging), and a workload score (homework, hours, heavy, tests), and overall sentiment
- *From ratings*: The difficulty rating, and whether a "would take again" value is present
- *Context*: Department grouped into STEM, social science, and humanities, school, and year of the review

Descriptive part:

- How ratings, difficulty, leniency, and quality scores are distributed
- Differences between department. For example, are STEM professors rated as harder and lower?
- Trends over time
- The most common words in high and low reviews

Explanatory part: Regress the quality rating on the leniency score and the teaching quality score, controlling for department and difficulty. Compare the size of the two coefficients.

Clustering: cluster professors on their average leniency, quality, difficulty, and rating. Can expect groups like "easy and liked" and "hard and disliked," or clustering by department.

Classification: predict whether a review is positive (rating 4 or higher) from its text features. Compare models and look at feature importance to determine which language drives positive ratings.

Hypotheses:

- *Leniency*
  - H1: More lenient professors receive higher ratings
  - H2: Higher difficulty ratings are associated with lower overall ratings
- *Teaching quality*
  - H3: Teaching quality has a smaller effect on ratings than leniency
- *Context*
  - H4: STEM professors receive lower ratings than humanities professors with similar leniency